

Objective:

This 4th year B.Techs course deals with the essential physical and mathematical processes involved in Quantum computation and quantum communication. is an introduction to the emerging field of Quantum Computers. Students are expected to know Quantum Mechanics but it is not mandatory. The phenomenon of entangled states and its use in controlled operations are introduced. With this basic tools from the physics of Quantum Mechanics the course intends to prepare the students to understand some of the important algorithms in quantum computation, quantum Information and Cryptography.

Prerequisite: Quantum Mechanics, Linear Algebra, Probability

1. Quantum Theory

- (i) Qubits
- (ii) Quantum Evolution
- (iii) Measurement Projective and POVM
- (iv) Composite system
- (v) Entanglement, Teleportation, Quantum Key

2. Noisy Quantum Theory

- (i) The density operator
- (ii) Composite quantum system
- (iii) Partial trace and reduced density operator
- (iv) Schmidt decomposition and Purification
- (v) EPR paradox and Bell inequalities.

3. Quantum Evolution

- (i) Evolution of a pure state
- (ii) Noisy system
- (iii) Noisy system evolution
- (iv) Bit flip and phase flip channel
- (v) Depolarizing channel
- (vi) Amplitude damping
- (vii) Phase damping

4. Quantum Error Correction

- (i) Three qubit bit flip code
- (ii) Three qubit phase flip code

(iii) The Shor code

5. Entanglement as a physical resource

- (i) Transforming bi-partite pure state entanglement
- (ii) Local operation and classical communication(LOCC).
- (iii) Entanglement distillation and dilution

Books:

1. Quantum Computation and Quantum Information
Michael A. Nielsen and Isaac L. Chuang
2. Quantum Information Theory
Mark M. Wilde.
3. An Introduction to Quantum Computing
Phillip Kaye, Raymond Laflamme and Michele Mosca
4. Modern Quantum Mechanics
J. J. Sakurai

Course material is based on Reference no. 1 . Reference 2 has much more than the course needs.
For introduction to basic ideas of Quantum Mechanics, J. J. Sakurai is good.
Evaluation: Two insem 20% each and endsem 50 %
Attendance : 10 %